How a macrophage *destroys* what it engulfs

Cell Biology seminar · First-year Medicine

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Phagocytic route · Enzyme route · Membrane fusion
Recycling · Pathological effect · Inflammatory signal

1 Alveolar macrophage

bronchiole
alveolar sac
this scene
tissue ~200 µm
cell ~20 µm
molecules ~10 nm

5.1 uptake
cleared ↑

2 Recognition

CD14
TLR4–MD-2
LPS
MyD88
NF-κB → inflammation
glycans
MR
iC3b
CR3
IgG
FcγR
Src
ITAM
Syk
PI3K
Cdc42 → WASP
Rac → WAVE
Arp2/3
F-actin
10 nm
bacterium
pseudopod

2.1
2.2 Pseudopod

4 Fusion

a tether *b* zipper *c* pore *d* recycle
syntaxin-7 Vti1b syntaxin-8
Rab7
HOPS
Arl8b
N → C
VAMP7
NSF
α-SNAP

2.7 Killing

ROS
NADPH
NOX2
e⁻
$O_2 \rightarrow O_2^{\bullet-} \rightarrow H_2O_2$
acid hydrolases
V-ATPase
H^+
reuse
lysozyme
cathepsins
lipases
nucleases
pH 7 6.5 6 5.5 5 4.5

Nucleus

3.1 Rough ER
COPII
cis
3.2 Golgi
trans
3.3 TGN
3.4 Clathrin vesicle
Rab7
3.5 Late endosome
2.3 Nascent phagosome
Early endosome
Rab5-GTP
2.4 Early phagosome
pH 6.5
Rab7-GTP
2.5 Late phagosome
pH 5.5
4
3.6 Lysosome
2.6 Phagolysosome
pH 4.5
recycled

3 Lysosome biogenesis

mRNA
Sec61
Glc
BiP
3.1 Rough ER
COPII
GlcNAc mannose glucose phosphate
Lys patch
phosphotransferase
3.2 cis-Golgi
3.3 TGN
3.4 Clathrin bud
uncovering enzyme
M6P
clathrin AP-1 · GGA
V-ATPase
M6PR
3.5 Late endosome
LAMP
3.6 Lysosome

5 Particles

5.2 Silica
5.2 Silica
NLRP3
engulfed
crystal
ROS
cathepsin B
K⁺
outside
signal 1
TLR → NF-κB
signal 2
caspase-1
NLRP3
inflammasome
ASC
IL-1β
IL-18
GSDMD pore
~20 nm
pyroptosis?
re-uptake
IL-1β · TNF-α
open cup
5.3 Asbestos
> 15–20 µm
ROS · enzymes
TGF-β · PDGF
collagen
5.4 Inflammation
5.5 Fibrosis

free
type I
type II
capillary

6 *Two routes converge to destroy; when the cargo persists, the same machinery injures the lung.*

Original illustration · schematic, not to scale

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